Linear Representations and Equations — Telling Slope and y-Intercept Apart

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 17	3. 2	5. 9	7. -4, 54
2. 2	4. 4	6. 8, 8	8. —

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.B.5, 8.EE.B.6 – *problems 1, 2, 3, 6, 7*
8.F.A / 8.F.B – *problems 4, 5, 8*
Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

1. *If they got 23: swapped the roles of the two numbers, multiplying by the constant term and adding the coefficient*
2. *If they got 58: took the leading number as the slope and the coefficient of x as the y-intercept because of the order they were written in*
3. *If they got 0.5: divided the horizontal change by the vertical change, which inverts the slope*
4. *If they got 13: reported the first number in the y row as the rate of change instead of finding how much y changes per step*
5. *If they got 13: called the first y value in the table the y-intercept, but that row is at x equal to one and not at x equal to zero*
6. *If they got 20: used the y coordinate of the given point as the y-intercept, but that point is not on the y axis*
7. *If they got -0.25: divided the change in time by the change in gallons, which inverts the rate of change*
7. *If they got 46: reported the first measured amount as the starting amount, but that reading was taken after the draining had already begun*

1. Find and fix: Lin reads $y = 3x + 5$.

(a) Step 1: name the slip. The slope is the number *attached to* x, so $m = 3$; the y-intercept is the number standing alone, so $b = 5$. Lin matched them by position on the page rather than by role.

(b) Step 2: substitute into the equation as written: $y = 3(4) + 5 = 12 + 5$.

Step 3: $y = \boxed{17}$.

Step 4: check the roles with the $x = 0$ test — $y = 3(0) + 5 = 5$, so 5 is the value at $x = 0$, which is exactly what a y-intercept is ✓. And each step of 1 in x raises y by 3, which is what a slope is.

Watch for: 23, from $5(4) + 3$. The swap changes the answer because x is multiplied by the wrong number.

2. Find and fix: Rae reads $y = 12 - 2x$.

(a) Step 1: the order the terms are written in carries no meaning — $12 - 2x$ and $-2x + 12$ are the same expression. What identifies the slope is that it multiplies x, so $m = -2$, and the constant 12 stands alone, so $b = 12$.

(b) Step 2: substitute: $y = 12 + (-2)(5) = 12 - 10$.

Step 3: $y = \boxed{2}$.

Step 4: check by direction — the slope is negative, so y must *fall* as x grows. From $x = 0$ (where $y = 12$) to $x = 5$, y drops to 2 ✓. Rae's answer of 58 is far above the starting value, which is impossible for a falling line.

Watch for: 58, from $-2 + 12(5)$. A wrong answer larger than the value at $x = 0$ on a decreasing line is a role swap, not an arithmetic slip.

3. Find and fix: Dev's slope through $(2, 3)$ and $(6, 11)$.

(a) Step 1: Dev put the change in x on top. Slope answers "how much does y change per 1 of x", so the change in y goes on top: $m = \dfrac{\text{change in } y}{\text{change in } x}$.

Step 2: compute in that order: $m = \dfrac{11 - 3}{6 - 2} = \dfrac{8}{4}$.

(b) Step 3: the slope is $\boxed{2}$.

Step 4: check by walking the line — from $x = 2$ to $x = 6$ is 4 steps, and 4 steps at a slope of 2 raises y by 8, taking 3 up to 11 ✓.

Watch for: 0.5, the reciprocal. An inverted slope and a correct one are always reciprocals, so if your answer times the right answer gives 1, you divided the wrong way round.

4. Find and fix: Sofia calls 13 the rate of change.

(a) Step 1: a rate of change measures how much y moves for each step of 1 in x. It is a *difference between rows*, not a number read out of one row. The value 13 is a single output, so it can never be a rate.

Step 2: subtract consecutive y values: $17 - 13 = 4$, and check it repeats: $21 - 17 = 4$ and $25 - 21 = 4$.

(b) Step 3: y increases by $\boxed{4}$ for every step of 1 in x.

Step 4: check that the x steps really are 1 apart — $1, 2, 3, 4$ ✓. If they were not, each difference would have to be divided by its own x step.

Watch for: 13. Any "rate" taken from a single cell of a table is this error.

5. Find and fix: Sofia calls 13 the y-intercept.

(a) Step 1: the y-intercept is the output at $x = 0$. This table starts at $x = 1$, so the intercept is not in the table at all — 13 is the value at $x = 1$, one step to the right of where the intercept lives.
(b) Step 2: walk one step backwards. Going left by 1 in x lowers y by the step size, so $13 - 4$.
Step 3: the y-intercept is $\boxed{9}$.
Step 4: check by rebuilding the rule: $y = 4x+9$ gives $4(1)+9 = 13$ at $x = 1$ and $4(4)+9 = 25$ at $x = 4$, matching both ends of the table ✓.
Watch for: 13. A table that does not contain $x = 0$ cannot show you the intercept — you always have to step back to it.

6. Find and fix: Owen's line, slope 4 through $(3, 20)$.

(a) Step 1: test Owen's equation $y = 4x + 20$ at $x = 3$: $4(3) + 20 = 32$, not 20. So his line misses the point it was supposed to pass through. The y-value of a point is only the intercept when that point sits on the y-axis, i.e. when $x = 0$ — and here $x = 3$.
(b) Step 2: put the point into $y = 4x + b$ and solve for b: $4(3) + b = 20$, so $12 + b = 20$.
Step 3: $b = \boxed{8}$, so the correct line is $y = 4x + 8$.
Step 4: check the point again: $4(3) + 8 = 20$ ✓.
Watch for: 20. Whenever a written equation fails the very point it was built from, suspect that a coordinate was used as an intercept.

7. Find and fix: the draining pool.

(a) Step 1: both errors come from confusing a rate with a reading. The rate of change is gallons *per minute*, so gallons go on top: $m = \dfrac{22 - 46}{8 - 2} = \dfrac{-24}{6}$.
Step 2: the rate of change is $\boxed{-4}$ gallons per minute — the pool loses 4 gallons each minute.
(b) Step 3: the starting amount is the value at 0 minutes, and the first reading was taken at 2 minutes, after 8 gallons had already drained. Use $b = y - mx$ with the point $(2, 46)$: $b = 46 - (-4)(2) = 46 + 8$.
Step 4: the pool held $\boxed{54}$ gallons at the start.
Step 5: check with the other reading: $-4(8) + 54 = -32 + 54 = 22$ ✓, which is the 8-minute measurement.
Watch for: -0.25 (the inverted rate) and 46 (a reading treated as a starting value). Notice the two errors are the same mistake twice: a number that belongs to one particular moment was used as though it described the whole relationship.

8. Explain: two different questions.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.

(a) The slope answers "for each increase of 1 in x, how much does y change?" It compares two quantities, so it is a rate and carries units of y per unit of x. The y-intercept answers "what is y when x is 0?" That is one output at one input, so it is a single value in the units of y alone. A rate describes the whole line; the intercept describes one point on it.

(b) In $y = mx + b$ the two numbers sit in structurally different places: m is multiplied by x and b is not. Swapping them changes both the steepness and the starting height. For example $y = 3x + 5$ passes through $(0, 5)$ climbing 3 per step, while $y = 5x + 3$ passes through $(0, 3)$ climbing 5 per step — different height, different tilt, and they agree at only one point. The same digits do not make the same line, because position in the equation *is* the meaning.

(c) *Table:* the intercept is the y value in the row where $x = 0$; if that row is missing, find the step size first and walk backwards to it. The slope is the change in y divided by the change in x between any two rows. *Graph:* the intercept is where the line crosses the y-axis; the slope is rise over run between two lattice points. *Equation with the constant first:* ignore the order and look at the roles — the number multiplied by x is the slope, the number standing alone is the intercept.

What a full-credit answer must contain: slope named as a rate per unit of x and the intercept as the output at $x = 0$ in (a); a concrete demonstration that swapping produces a different line in (b); and in (c) a method for each of the three representations, including what to do when a table omits $x = 0$.